

RenovaTrack

Software Requirements Document

Renovation Project Cost Tracker

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Tech Stack: Next.js · Supabase · Tailwind CSS

Prepared by WebCross for Developer Handoff

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1. Project Overview

Background

The client is a property developer who currently manages renovation project costs across multiple Excel spreadsheets. Each project has two trackers: a week-by-week granular expense log and a high-level budget vs. actual summary. The goal is to replace both with a single web application that works on mobile (on-site) and desktop.

Goals

- Replace Excel-based tracking with a fast, mobile-friendly web app
- Support multiple renovation projects simultaneously
- Auto-calculate labour costs, materials costs, and VAT
- Track payments per trade and supplier with payment status
- Provide a real-time dashboard with budget vs. forecast variance
- Allow receipt / invoice photo uploads attached to each expense entry
- Export data as PDF or Excel for accountants and stakeholders

Out of Scope (v1)

- Client-facing portal
- Invoice generation / payment collection from clients
- Multi-user team collaboration (single user per account in v1)
- Third-party accounting integrations (Xero, QuickBooks)

2. Tech Stack

Layer	Technology
Framework	Next.js 14 (App Router)
Language	TypeScript
Styling	Tailwind CSS
Database	Supabase (PostgreSQL)
Auth	Supabase Auth — email/password
File Storage	Supabase Storage
ORM / Query	Supabase JS Client v2
PDF Export	@react-pdf/renderer
Excel Export	xlsx (SheetJS)
Deployment	Vercel

Folder Structure

```
/app
/dashboard → Dashboard page
/projects → Project list + create
/projects/[id] → Project detail (tabs: overview, expenses, trades, materials)
/projects/[id]/expenses/new → Add expense form
/settings → Trade rate lookups
/api → Route handlers
/components
/ui → Reusable UI components
/forms → Form components
/charts → Chart components
/lib
/supabase.ts → Supabase client
/calculations.ts → Cost calculation helpers
/export.ts → PDF/Excel export helpers
/types
index.ts → Shared TypeScript types
```

3. User Roles & Authentication

v1 Auth Model

Single-user account. One email/password login per account. All data is private to the logged-in user, enforced via Supabase Row-Level Security (RLS).

Supabase Auth Setup

Setting	Value
Provider	Email + Password
Email confirmation	Enabled
Redirect after login	/dashboard
Password reset	Enabled via Supabase default flow

Row-Level Security (RLS)

Every table must include a **user_id uuid** column referencing auth.users(id). All RLS policies follow this pattern:

```
CREATE POLICY "Users can only access their own data"
ON projects FOR ALL
USING (auth.uid() = user_id);
```

4. Database Schema

4.1 projects table

Represents a single renovation property / project.

Column	Type	Notes
id	uuid PK	gen_random_uuid()
user_id	uuid FK	References auth.users — RLS key
name	text NOT NULL	e.g. '46 Glenferrie Rd'
address	text	Full property address
target_budget	numeric(12,2)	Client's budget ceiling, default 0
contingency_pct	numeric(5,2)	e.g. 10 for 10%, default 0
start_date	date	Optional
end_date	date	Optional
status	text	active completed paused

Column	Type	Notes
notes	text	Free-text notes
created_at	timestamptz	Default now()
updated_at	timestamptz	Default now()

4.2 expense_entries table

Core table. Each row is one line-item expense. This replaces every row in the client's Week-by-Week Plan sheet.

Column	Type	Notes
id	uuid PK	gen_random_uuid()
user_id	uuid FK	RLS key
project_id	uuid FK	References projects(id) CASCADE
week_number	integer	Week 1, 2, 3 ... required
description	text NOT NULL	Task / description
category	text	Labour Materials Skip/Disposal Other
trade	text	References trade_lookups.name
location_room	text	Optional
notes	text	Dependencies / extra info
supplier	text	Optional
invoice_ref	text	Invoice / receipt reference
paid_date	date	Optional
payment_method	text	Cash Debit Card Credit Card Bank Transfer
hours	numeric(8,2)	Labour hours, default 0
labour_rate	numeric(8,2)	£/hr, auto-filled from trade lookup
direct_labour_cost	numeric(12,2)	Used when hours not tracked
qty	numeric(10,2)	Materials quantity, default 0
unit_cost	numeric(12,2)	Materials unit cost
vat_rate	numeric(5,2)	0 or 20
status	text	Planned In Progress Paid Cancelled
receipt_url	text	Supabase Storage URL
created_at	timestamptz	Default now()
updated_at	timestamptz	Default now()

NOTE — total_incl_vat is NOT stored in the database. It is computed in the application layer on every read: $\text{base_cost} = (\text{hours} \times \text{rate OR direct_labour_cost}) + (\text{qty} \times \text{unit_cost})$, then $\text{total_incl_vat} = \text{base_cost} + (\text{base_cost} \times \text{vat_rate} / 100)$.

4.3 trade_lookups table

Column	Type	Notes
id	uuid PK	gen_random_uuid()
user_id	uuid FK	RLS key
name	text NOT NULL	e.g. 'General Builder'
default_rate	numeric(8,2)	£/hr default, editable by user
default_markup_pct	numeric(5,2)	Markup %, default 0
created_at	timestampz	Default now()

Default Seed Data (insert on first login)

Trade	Default Rate (£/hr)	Markup %
General Builder	45	10
Carpenter/Joiner	50	10
Plumber	60	10
Electrician	65	10
Plasterer	55	10
Tiler	55	10
Decorator	45	10
Roofer	60	10
Groundworker	55	10
Kitchen Fitter	55	10
Bathroom Fitter	55	10
Skip/Disposal	0	0
Other	50	10

4.4 project_weeks table (optional — completion tracking)

Column	Type	Notes
id	uuid PK	gen_random_uuid()
project_id	uuid FK	References projects(id) CASCADE
week_number	integer NOT NULL	Unique per project

Column	Type	Notes
completion_pct	numeric(5,2)	Manual input 0–100
notes	text	Optional week notes

4.5 Indexes

```
CREATE INDEX idx_expense_entries_project_id ON expense_entries(project_id);  
CREATE INDEX idx_expense_entries_user_id ON expense_entries(user_id);  
CREATE INDEX idx_expense_entries_week ON expense_entries(project_id, week_number);  
CREATE INDEX idx_expense_entries_status ON expense_entries(status);
```


5. Feature Modules

Module 1: Project Management

Create, edit, delete renovation projects. Fields: Name, Address, Target Budget, Contingency %, Start Date, End Date, Notes, Status (Active / Completed / Paused). On creation, seed trade_lookups with default trades. Delete cascades all expense entries after a confirmation dialog requiring the user to type the project name.

Module 2: Expense Entry (Core Feature)

The primary data entry screen — replaces the Week-by-Week Plan sheet. See Section 5.2 below for full field specification. Key behaviour: selecting a trade auto-fills the labour rate; category selection shows/hides relevant fields; all cost calculations update live as the user types.

Module 3: Expense List

Filterable, sortable table of all entries per project. Filters: week range, category, trade, status, payment method. Inline actions: Edit, Delete, Mark as Paid. Totals row at bottom: Labour, Materials, VAT, Grand Total.

Module 4: Trades & Labour Tracker

Aggregated view grouped by trade/person name. Shows: Quoted, Actual, Paid, Remaining, Status badge (Paid / Partial / Pending). Derived entirely from expense_entries — no separate table required.

Module 5: Materials & Suppliers Tracker

Aggregated view grouped by supplier. Shows all material purchases with cost, VAT, total, and payment method. Derived from expense_entries WHERE category = 'Materials'.

Module 6: Project Dashboard

Summary cards: Target Budget, Forecast Total, Variance (red if over, green if under), Contingency, Forecast + Contingency, Paid to Date, Remaining to Pay. Charts: weekly spend bar chart, category donut (Labour vs Materials). Week-by-week table with manual Completion % input.

Module 7: Export

PDF export — full project report with cover page, budget summary, week table, trades summary, materials summary. Uses @react-pdf/renderer. Excel export — recreates original sheet structure (Week-by-Week, Summary, Trades, Materials). Uses SheetJS (xlsx).

Module 8: Trade Lookups / Settings

Manage default trade rates. List, inline-edit, add, delete. Deletion blocked if expense entries reference the trade.

5.2 Expense Entry — Full Field Specification

Field	Type	Required	Notes
Week Number	Integer	Yes	Auto-suggests next week from last entry
Description	Text	Yes	Max 200 chars
Category	Select	Yes	Labour Materials Skip/Disposal Other
Trade	Select	No	From trade_lookups — auto-fills rate
Location / Room	Text	No	e.g. Kitchen, Bathroom
Notes	Text	No	Dependencies, extra info
Supplier	Text	No	Company or person name
Invoice / Receipt Ref	Text	No	Reference number
Paid Date	Date	No	
Payment Method	Select	No	Cash Debit Card Credit Card Bank Transfer
Hours (Labour)	Decimal	No	Greys out if category = Materials
Labour Rate (£/hr)	Decimal	No	Auto-filled from trade, editable
Direct Labour Cost	Decimal	No	Used when hours not tracked
Qty (Materials)	Decimal	No	Greys out if category = Labour
Unit Cost (£)	Decimal	No	
VAT Rate	Select	Yes	0% or 20%
Status	Select	Yes	Planned In Progress Paid Cancelled
Receipt Upload	File	No	Image or PDF, max 10MB

6. Pages & UI Specifications

Route	Description
/ (Login)	Email + password. Forgot password link. No public sign-up in v1.
/dashboard	All projects as cards. Each shows name, address, budget used %, total spent, status badge. 'Create project' CTA.
/projects/new	Create project form. Cancel returns to dashboard.
/projects/[id]	Tabbed layout: Overview Expenses Trades & Labour Materials. Persistent top bar with project name, week indicator, budget % pill, Export dropdown.
/projects/[id] — Overview tab	Budget overview cards, weekly spend bar chart, category breakdown, week-by-week table with completion % input.
/projects/[id] — Expenses tab	Filter bar + expense table + Add Expense button. Totals row at bottom. On desktop: slide-over drawer (700px). On mobile: full-screen form.
/projects/[id] — Trades tab	Grouped trades table with Quoted / Paid / Remaining / Status per trade.
/projects/[id] — Materials tab	Grouped materials by supplier with totals.
/settings	Trade lookup table. Inline edit, add row, delete row.

Mobile requirement: The client uses the app on-site on a phone. All forms must be fully usable at 375px viewport. Minimum touch target size 44px. Consider a bottom navigation bar on mobile (Dashboard, Add Expense, Projects, Settings).

7. API Routes

All routes are Next.js Route Handlers under `/app/api/` using the Supabase server client. All routes require a valid session.

Projects

Method	Route	Description
GET	/api/projects	List all projects for current user
POST	/api/projects	Create new project
GET	/api/projects/[id]	Get single project details
PATCH	/api/projects/[id]	Update project fields
DELETE	/api/projects/[id]	Delete project + cascade all entries

Expenses

Method	Route	Description
GET	/api/projects/[id]/expenses	List all expenses (filter via query params)

Method	Route	Description
POST	/api/projects/[id]/expenses	Create expense entry
GET	/api/projects/[id]/expenses/[eid]	Get single entry
PATCH	/api/projects/[id]/expenses/[eid]	Update entry
DELETE	/api/projects/[id]/expenses/[eid]	Delete entry

Expense Filter Query Params:

Param	Example	Effect
week	?week=5	Filter by single week
week_from / week_to	?week_from=1&week_to=10	Filter by week range
category	?category=Labour	Filter by category
status	?status=Paid	Filter by status
trade	?trade=Plasterer	Filter by trade

Aggregations & Summary

Method	Route	Description
GET	/api/projects/[id]/summary	Budget overview stats
GET	/api/projects/[id]/summary/by-week	Array of weekly totals
GET	/api/projects/[id]/summary/by-category	Labour vs materials breakdown
GET	/api/projects/[id]/trades	Aggregated trades payment summary
GET	/api/projects/[id]/materials	Aggregated materials/suppliers

Trade Lookups & Export

Method	Route	Description
GET	/api/lookups/trades	Get all trade types and rates
POST	/api/lookups/trades	Add new trade
PATCH	/api/lookups/trades/[id]	Update trade rate
DELETE	/api/lookups/trades/[id]	Delete trade
POST	/api/expenses/[eid]/receipt	Upload receipt to Supabase Storage
DELETE	/api/expenses/[eid]/receipt	Delete receipt from storage
GET	/api/projects/[id]/export/pdf	Generate and return PDF
GET	/api/projects/[id]/export/excel	Generate and return .xlsx file

8. Business Logic & Calculations

Cost Calculation (TypeScript — lib/calculations.ts)

```
export function calcLabourCost(hours: number, rate: number, directCost: number): number {
  if (hours > 0 && rate > 0) return hours * rate;
  return directCost ?? 0;
}

export function calcMaterialsCost(qty: number, unitCost: number): number {
  if (qty > 0 && unitCost > 0) return qty * unitCost;
  return 0;
}

export function calcTotal(labourCost: number, materialsCost: number, vatRate: number) {
  const subtotal = labourCost + materialsCost;
  const vatAmount = subtotal * (vatRate / 100);
  return { subtotal, vatAmount, totalInclVat: subtotal + vatAmount };
}
```

Calculation Rules

Formula	Expression
Labour Cost	hours × labour_rate OR direct_labour_cost (when hours = 0)
Materials Cost	qty × unit_cost
Subtotal	Labour Cost + Materials Cost
VAT Amount	Subtotal × (vat_rate ÷ 100)
Total incl. VAT	Subtotal + VAT Amount
Forecast Total	Sum of total_incl_vat for all non-Cancelled entries
Paid to Date	Sum of total_incl_vat for status = 'Paid' entries
Variance	Forecast Total – Target Budget
Contingency Amount	Target Budget × (contingency_pct ÷ 100)
Forecast + Cont.	Forecast Total + Contingency Amount

Display Rules

- Variance positive (over budget) → display in red
- Variance negative (under budget) → display in green
- All amounts displayed to 2 decimal places with £ prefix
- Week number auto-suggests: increment from the last used week number on this project
- VAT options in v1: 0% and 20% only
- Calculations must run live client-side as the user types (before saving)

9. File Storage & Uploads

Setting	Value
Bucket name	receipts
Access	Private — authenticated requests only
Max file size	10 MB
Allowed types	image/jpeg, image/png, image/webp, application/pdf
File naming	receipts/{user_id}/{project_id}/{expense_id}/{timestamp}-{filename}

Upload Flow

1. User selects file in the expense form
2. Client uploads directly to Supabase Storage using a signed upload URL
3. On successful upload, save the storage path to expense_entries.receipt_url
4. Display a paperclip icon in the expense row if a receipt exists — click opens in new tab
5. In the edit form, show thumbnail preview (image) or PDF icon with option to delete and re-upload

10. Non-Functional Requirements

Performance

Metric	Target
Initial page load	< 2 seconds on 4G mobile
Form submission round-trip	< 500 ms
Dashboard charts render	< 1 second after data load
PDF / Excel export generation	< 5 seconds

Data Validation

All validation must run both client-side (instant feedback) and server-side (API layer).

Field	Rule
Week number	Positive integer, required
Description	Required, max 200 characters
Hours	Non-negative decimal, max 999
Labour rate	Non-negative decimal
Qty	Non-negative decimal
Unit cost	Non-negative decimal
VAT rate	Must be exactly 0 or 20
Target budget	Non-negative number

Security

- All API routes protected by Supabase Auth — check session on every request
- RLS enabled on ALL tables — users cannot access other users' data
- File uploads: validate MIME type server-side before saving URL to database
- No sensitive data in URL query params
- Use Supabase server client (not browser client) in Route Handlers

Error Handling

- Show user-friendly toast notifications for all errors
- Form validation errors shown inline (below the relevant field)
- If API call fails, do not clear the form — allow user to retry
- 404 page for unknown project IDs
- Loading skeleton states for all data-fetching screens
- Empty states for projects list, expenses list, etc.

11. Implementation Phases

Phase 1 — Foundation

Weeks 1–2

- Project setup: Next.js 14, TypeScript, Tailwind CSS, Supabase
- Supabase: create all tables, RLS policies, indexes
- Auth: login page, session middleware, protected routes
- Trade lookups: seed data + Settings page
- Project CRUD: create, list, edit, delete

Phase 2 — Core Expense Tracking

Weeks 3–4

- Add Expense form with all fields + live cost calculations
- Expense list per project with filters
- Edit / delete expense entry
- Quick 'Mark as Paid' action
- Receipt file upload (Supabase Storage)

Phase 3 — Dashboard & Analytics

Week 5

- Project Overview tab: budget cards, weekly spend bar chart
- Category breakdown chart (Labour vs Materials)
- Week-by-week summary table with Completion % input
- Trades & Labour aggregated view
- Materials & Suppliers aggregated view

Phase 4 — Export & Polish

Week 6

- PDF export (@react-pdf/renderer)
- Excel export (SheetJS)
- Mobile layout and bottom nav
- Loading states, error states, empty states
- Final QA and bug fixes

Appendix A — Example Data

Extracted from the client's existing Excel files. Use as seed/test data during development.

We ek	Description	Category	Trade	Cost (£)
1	Labourers	Labour	General Builder	450
1	DMC - Dave	Labour	General Builder	220
1	Adam - Clearance	Labour	Skip/Disposal	1,100
2	Labourers	Labour	General Builder	675
2	Adam - Clearance	Labour	Skip/Disposal	1,000
5	Skip (Stevenage Skips)	Labour	Skip/Disposal	480
5	Dave Barker Groundwork	Labour	General Builder	1,250
5	Master Mix	Materials	Other	1,025
6	Steels	Materials	General Builder	1,000
9	Owen Brickwork	Labour	General Builder	1,935
9	Dave Builder (steels in)	Labour	General Builder	2,035
9	Toby (earth clearance)	Labour	Skip/Disposal	1,240
15	Various (largest week)	Mixed	Mixed	7,893

Project Totals (from existing data):

Metric	Value
Forecast Total	£42,411
Weeks tracked	15
Largest single week	Week 15 — £7,893
Labour total	~£39,000 (estimated)
Materials total	~£3,400 (estimated)

Appendix B — Glossary

Term	Definition
Labour Cost	hours × labour_rate OR direct_labour_cost when hours not tracked
Materials Cost	qty × unit_cost
Subtotal	Labour Cost + Materials Cost for one entry
Total incl. VAT	Subtotal + (Subtotal × vat_rate%)

Term	Definition
Forecast Total	Sum of all non-Cancelled entries' total_incl_vat
Target Budget	The project's intended spend ceiling set by the client
Variance	Forecast Total minus Target Budget
Contingency	Additional buffer percentage on top of the target budget
Trade	Type of tradesperson: Builder, Plumber, Electrician, etc.
Status: Planned	Entry logged but work not yet started or paid
Status: In Progress	Work underway, payment not yet made
Status: Paid	Payment confirmed and made
Status: Cancelled	Entry excluded from all forecast calculations
RLS	Row-Level Security — Supabase/Postgres policy restricting data access per user

Document prepared by WebCross | For questions on requirements, contact the project owner.